

DOCKER SWARM Practical

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PS C:\Users\charl> docker network create net1
ed50ed78bd7ce1a22bf857fb30603a9d6fd2c93da8320516e880e6751e043f4d
PS C:\Users\charl> docker network ls\
docker: unknown command: docker network ls\

Usage:  docker network

Run 'docker network --help' for more information
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PS C:\Users\charl> docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
7f5bf6d830df        bridge             bridge              local
026c95e8d1bf        compose1_default   bridge              local
d6dcfa1d69f1        compose3_default   bridge              local
3cc4b96fd102        docker-compose-3_default bridge              local
b5ee194955e9        docker-compose-4_appnetwork bridge              local
85dc12244867        dockercompose1_default bridge              local
b4b94023171e        host               host                local
8ce6912c710d        l1                 bridge              local
a3efba63fc8c        mynetwork          bridge              local
4175c03ba60d        mynetwork1         bridge              local
ed50ed78bd7c        net1               bridge              local
b91c16406d2c        node-mongo-compose1_default bridge              local
1049be734fa8        none               null                local
0683a620b45e        smart-log-analyzer_default bridge              local
fd833cc1fbbb        wordpress_docker_default bridge              local
PS C:\Users\charl> docker run -dit --name manager -p 8085:8080 --privileged --network net1 docker:dind
unknown flag: --privileged

Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Run 'docker run --help' for more information
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PS C:\Users\charl> docker run -dit --name manager -p 8085:8080 --privileged --network net1 docker:dind
Unable to find image 'docker:dind' locally
dind: Pulling from library/docker
fa2c3644d420: Pull complete
6a0ac1617861: Pull complete
5b2d5a3c6758: Pull complete
8d983ea1647a: Pull complete
af53c3ea4133: Pull complete
90e4e9c4f171: Pull complete
4f4fb700ef54: Pull complete
a7f5a18659b1: Pull complete
6642357ac7fa: Pull complete
c1aad62b3176: Pull complete
042a8c82cc77: Pull complete
a5e52c67d040: Pull complete
214b558d88b0: Pull complete
c6c9e6578d78: Pull complete
704fb81a37c3: Pull complete
e202d21d268c: Pull complete
Digest: sha256:c77e5d7912f9b137cc67051fdc2991d8f5ae22c55ddf532bb836dcb693a04940
Status: Downloaded newer image for docker:dind
22802cd05e4ff3e8e3c12bbb52d7e23d79cbf7e03460f4d0d192332753449574
PS C:\Users\charl> docker run -dit --name worker1 --privileged --network swarm-net docker:dind
8e37d02c920fbf6fe61581cbd9b381c929895e2c7a498d1494bfe5e8a3db8ec1
```

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Run 'docker run --help' for more information
PS C:\Users\charl> docker run -dit --name worker1 --privileged --network swarm-net docker:dind
docker: Error response from daemon: Conflict. The container name "/worker1" is already in use by container "8e37d02c920fbffe61581cbd9b381c9298e2c7a498d1494bfbfe5e8a3db8ec1". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\Users\charl> docker run -dit --name worker2 --privileged --network swarm-net docker:dind
7be1a4e74e4844abcbd89ac6ae0e393f383b9b84355fc0c8bcd5248685ac9890
docker: Error response from daemon: failed to set up container networking: network swarm-net not found

Run 'docker run --help' for more information
PS C:\Users\charl> docker exec -it manager sh
/ # docker swarm init
Swarm initialized: current node (3y2aak5juhzfuwt73mjvnrriiv) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-2og4kid8xqmk04vfv2c2u9ifneq7us5kruio9xzpt5m9v5k2r-2djrrnwxqok8tvz4tonjclxou 172.29.0.2:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
/ # exit

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PS C:\Users\charl> docker network create swarm-net
96d3722b6b43cb6c7ecf403a8e3491173413a7ad20ed7dae04821a251b8fbb9d
PS C:\Users\charl> docker run -dit --name worker1 --privileged --network swarm-net docker:dind dockerd
54a4ae52d7e79fb311db8666269f014407c2e763a2a5a27271eb452de15e3fee
PS C:\Users\charl> docker run -dit --name worker2 --privileged --network swarm-net docker:dind dockerd
cddc3293fdc2f639ce9b48101d3494151eac8aff5b2be1dde6214335502660bb
PS C:\Users\charl> docker exec -it worker1 sh
/ # docker swarm join --token SWMTKN-1-2og4kid8xqmk04vfv2c2u9ifneq7us5kruio9xzpt5m9v5k2r-2djrrnwxqok8tvz4tonjclxou 172.29.0.2:2377
Error response from daemon: Timeout was reached before node joined. The attempt to join the swarm will continue in the background. Use the "docker info" command to see the current swarm status of your node.
/ # docker swarm join --token SWMTKN-1-2og4kid8xqmk04vfv2c2u9ifneq7us5kruio9xzpt5m9v5k2r-2djrrnwxqok8tvz4tonjclxou 172.29.0.2:2377
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
/ # docker swarm leave
Node left the swarm.
/ # docker swarm join --token SWMTKN-1-2og4kid8xqmk04vfv2c2u9ifneq7us5kruio9xzpt5m9v5k2r-2djrrnwxqok8tvz4tonjclxou 172.29.0.2:2377
Error response from daemon: Timeout was reached before node joined. The attempt to join the swarm will continue in the background. Use the "docker info" command to see the current swarm status of your node.
/ # exit
PS C:\Users\charl> docker exec -it worker2 sh
/ # docker swarm join --token SWMTKN-1-2og4kid8xqmk04vfv2c2u9ifneq7us5kruio9xzpt5m9v5k2r-2djrrnwxqok8tvz4tonjclxou 172.29.0.2:2377
Error response from daemon: Timeout was reached before node joined. The attempt to join the swarm will continue in the background. Use the "docker info" command to see the current swarm status of your node.
/ # exit

```

```

/ # exit
PS C:\Users\charl> docker exec -it manager sh
/ # docker service create --name web1 --replicas 4 -p 8085:80 nginx
m8fkiduh03d1fzfu3hbjug2he
overall progress: 4 out of 4 tasks
1/4: running [=====>]
2/4: running [=====>]
3/4: running [=====>]
4/4: running [=====>]
verify: Service m8fkiduh03d1fzfu3hbjug2he converged
/ # docker service ps web
no such service: web
/ # docker service ps web1

```

ID	NAME	IMAGE	NODE	DESIRED STATE	CURRENT STATE	ERROR	PORTS
otd67t4ogp4h	web1.1	nginx:latest	22802cd05e4f	Running	Running about a minute ago		
lyzxle3o9ags	web1.2	nginx:latest	22802cd05e4f	Running	Running about a minute ago		
3o9cf8jjk1t9	web1.3	nginx:latest	22802cd05e4f	Running	Running about a minute ago		
gy2m37n9bhan	web1.4	nginx:latest	22802cd05e4f	Running	Running about a minute ago		

```

/ # docker node ls

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ID	HOSTNAME	STATUS	AVAILABILITY	MANAGER STATUS	ENGINE VERSION
3y2aak5juhzfuwt73mjvnrriiv *	22802cd05e4f	Ready	Active	Leader	29.4.1